Light

# GCSE Computer Science – Networks Basics (Set 2)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the main purpose of connecting computers together in a network?

- ○ **A.** To allow computers to share data and resources with each other
- ○ **B.** To make each computer run faster
- ○ **C.** To increase the storage capacity of each computer
- ○ **D.** To reduce the electricity used by computers

Q2 (1 mark)

Who typically owns and manages a Local Area Network (LAN)?

- ○ **A.** The government

- ○ **B.** Multiple internet service providers working together
- ○ **C.** The general public
- ○ **D.** A single person or organisation

Q3 (1 mark)

How is the ownership of a Wide Area Network (WAN) typically different from a LAN?

- ○ **A.** WANs are often under collective or distributed ownership rather than a single owner
- ○ **B.** WANs are always owned by schools
- ○ **C.** WANs must be owned by the government
- ○ **D.** WANs have no owners at all

Q4 (1 mark)

What is the typical maximum range of a Bluetooth connection used in a Personal Area Network (PAN)?

- ○ **A.** Several kilometres
- ○ **B.** Across an entire building
- ○ **C.** Worldwide via satellite
- ○ **D.** About 10 metres

Q5 (1 mark)

What is one disadvantage of connecting computers to a network?

- ○ **A.** Files can be shared between computers
- ○ **B.** Printers can be shared by multiple users
- ○ **C.** Users can communicate more easily
- ○ **D.** Security risks increase as hackers could potentially access connected computers

Q6 (1 mark)

Why are network protocols necessary?

- ○ **A.** To ensure devices can communicate by following agreed rules for data transmission
- ○ **B.** To make cables connect more securely
- ○ **C.** To increase the speed of computer processors
- ○ **D.** To reduce the cost of network equipment

Q7 (1 mark)

What is the purpose of SMTP?

- ○ **A.** To browse websites securely
- ○ **B.** To download files from the internet
- ○ **C.** To connect Bluetooth devices

- ○ **D.** To send outgoing email messages

Q8 (1 mark)

Why would someone use IMAP to access their emails?

- ○ **A.** IMAP allows emails to be accessed from multiple devices while keeping them stored on the server
- ○ **B.** IMAP encrypts all web browsing
- ○ **C.** IMAP makes emails send faster
- ○ **D.** IMAP converts emails into web pages

Q9 (1 mark)

What does encryption do to protect data on a network?

- ○ **A.** It deletes data after it has been transmitted
- ○ **B.** It scrambles data so it cannot be read without the correct key
- ○ **C.** It speeds up data transmission
- ○ **D.** It compresses data to make files smaller

Q10 (1 mark)

What is the purpose of authentication in network security?

- ○ **A.** To verify the identity of a user or device before granting access

- B. To compress data before sending
- C. To increase network speed
- D. To convert wired signals to wireless

Q11 (1 mark)

Which is an advantage of wired networks compared to wireless networks?

- A. Wired networks allow users to move around freely
- B. Wired networks do not require any cables
- C. Wired networks have unlimited range
- D. Wired networks typically provide faster and more reliable connections

Q12 (1 mark)

At which layer of the TCP/IP model does the TCP protocol operate?

- A. Application layer
- B. Transport layer
- C. Internet layer
- D. Link layer

Q13 (2 marks)

What are the main functions of the transport layer in the TCP/IP model?

- ○ **A.** To display web pages and run email programs
- ○ **B.** To physically transmit signals through cables and wireless connections
- ○ **C.** To set up communication between hosts and agree settings such as packet size
- ○ **D.** To assign IP addresses to all devices on the network

Q14 (2 marks)

A data packet is being sent across the internet to a computer on the other side of the world. What role does the IP protocol play in this process?

- ○ **A.** IP adds source and destination addresses to the packet and determines the route it takes across networks
- ○ **B.** IP displays the data on the recipient's screen
- ○ **C.** IP encrypts the data so hackers cannot read it
- ○ **D.** IP physically transmits the electrical signals through the cables

Q15 (2 marks)

When downloading a large file, some data packets are lost during transmission. How does TCP handle this problem?

- ○ **A.** TCP ignores the lost packets and delivers an incomplete file
- ○ **B.** TCP cancels the entire download and requires the user to start again

○ **C.** TCP detects the missing packets and requests they be retransmitted until the file is complete

○ **D.** TCP automatically increases the internet speed to prevent future losses

Q16 (2 marks)

A network administrator configures a firewall with a rule to block all incoming traffic on port 23. What is the effect of this rule?

○ **A.** All computers on the network will run faster

○ **B.** The firewall will prevent any external connections attempting to use port 23 from entering the network

○ **C.** The firewall will encrypt all data on port 23

○ **D.** Users will no longer need passwords to access the network

Q17 (2 marks)

A café offers free Wi-Fi but wants to prevent non-customers from using it. The owner decides to use MAC address filtering. How would this work?

○ **A.** The café would require customers to create passwords before connecting

○ **B.** The café would encrypt all internet traffic automatically

○ **C.** The café would add customers' device MAC addresses to an approved list, only allowing those devices to connect

○ **D.** The café would limit the number of websites customers can visit

Q18 (2 marks)

An office using wireless networking notices that connection speeds drop significantly whenever the microwave in the break room is used. What explains this problem?

- ○ **A.** The microwave is downloading large files from the internet
- ○ **B.** Wireless signals can be disrupted by interference from electrical devices that emit electromagnetic radiation
- ○ **C.** The microwave is using too much electricity, reducing power to the network equipment
- ○ **D.** The wireless network needs more cables installed

Q19 (2 marks)

A factory floor has many heavy electrical machines that create electromagnetic interference. Which type of network cable would be most suitable for this environment?

- ○ **A.** Copper cables, because they are cheaper
- ○ **B.** Either type would work equally well in this environment
- ○ **C.** No cables would work; only wireless is suitable for factories
- ○ **D.** Fibre optic cables, because they transmit light and are immune to electromagnetic interference

Q20 (2 marks)

A business stores all its files on a central network server. One morning, the server crashes and employees cannot access their work. What network disadvantage does this illustrate?

- ○ **A.** Networks make it impossible to share files between computers
- ○ **B.** Networks always cost more than standalone computers
- ○ **C.** Dependency on central resources means a single failure can affect all users
- ○ **D.** Networks prevent employees from working together

Q21 (2 marks)

A student accesses a website using HTTP. At which layer of the TCP/IP model does HTTP operate, and what does this layer do?

- ○ **A.** Link layer—it handles the physical transmission of data through cables
- ○ **B.** Internet layer—it assigns IP addresses to all devices
- ○ **C.** Transport layer—it breaks data into packets and ensures delivery
- ○ **D.** Application layer—it is where network applications like web browsers operate

Q22 (2 marks)

What components and functions are found at the link layer of the TCP/IP model?

- ○ **A.** Web browsers and email programs that users interact with directly
- ○ **B.** Protocols that ensure reliable data delivery and retransmission of lost packets
- ○ **C.** Network hardware like network interface cards (NICs) and operating system device drivers

○ **D.** Protocols that determine how data packets are routed across the internet

Q23 (2 marks)

A bank has a network in its main building and another network in a branch 50 miles away. The main building network is classified as a LAN. Why would the connection between the two buildings be classified differently?

○ **A.** Because it uses different network cables

○ **B.** Because it is wireless instead of wired

○ **C.** Because it covers a wide geographical area between two separate locations, making it a WAN connection

○ **D.** Because it handles different types of data than the LAN

Q24 (3 marks)

A small accounting firm with 12 employees is considering whether to connect their computers in a network or keep them as standalone machines. Evaluate the advantages and disadvantages of creating a network for this firm.

○ **A.** They should not network because networks are only beneficial for large companies with hundreds of employees

○ **B.** The network would allow resource sharing and centralised backup, but creates dependency risks if the server fails and requires investment in security measures

○ **C.** They should network because it would eliminate all security risks and guarantee their data is always safe

○ **D.** Networking is only about printer sharing and has no other

significant benefits or drawbacks

Q25 (3 marks)

When a computer receives data from the internet, the data passes through the layers of the TCP/IP model in a specific order. Starting from the physical arrival of data, describe the correct order of layers.

- ○ **A.** Application → Transport → Internet → Link: data moves from user applications down to hardware
- ○ **B.** Link → Internet → Transport → Application: data arrives at hardware, is addressed, assembled, then delivered to applications
- ○ **C.** Transport → Link → Application → Internet: packets are assembled first, then physically transmitted
- ○ **D.** Internet → Application → Link → Transport: addressing happens first, then applications process it

Q26 (3 marks)

A dentist's surgery stores sensitive patient medical records digitally. The practice manager wants to implement comprehensive network security. Explain how combining authentication, encryption, and a firewall would protect the patient data.

- ○ **A.** Only encryption is needed because it makes all other security measures unnecessary
- ○ **B.** Authentication prevents unauthorised users, encryption protects data if intercepted, and a firewall blocks malicious network traffic—together they create multiple barriers against different types of threats
- ○ **C.** These three measures all do the same thing, so implementing one is sufficient
- ○ **D.** A firewall alone provides complete protection and the other

measures add no value

Q27 (3 marks)

A delivery driver uses a Bluetooth earpiece connected to their phone, the phone connects to the company's warehouse Wi-Fi to download delivery routes, and the warehouse system retrieves customer addresses from the company's headquarters database in another city. Identify the three types of networks involved.

- ○ **A.** Three WANs: all connections involve moving data between locations
- ○ **B.** Three LANs: all connections are owned by the same company
- ○ **C.** PAN (Bluetooth earpiece to phone), LAN (phone to warehouse Wi-Fi), WAN (warehouse to headquarters in another city)
- ○ **D.** Three PANs: all connections involve personal devices

Q28 (3 marks)

A new media production company is installing network cabling in their studio. They will be transferring very large video files (often several gigabytes each) between editing workstations. They are also located next to a radio transmitter. Evaluate the choice between fibre optic and copper cabling.

- ○ **A.** Copper would be better because it is more flexible and easier to install around corners
- ○ **B.** Either would work equally well since all network cables transfer data at the same speed
- ○ **C.** Fibre optic would be better because it provides higher bandwidth for large file transfers and is immune to interference from the nearby radio transmitter

○ **D.** They should use wireless networking instead of any cables

Q29 (3 marks)

A user is making an online purchase on a public Wi-Fi network at an airport. The website uses HTTPS. Explain why HTTPS is particularly important in this situation and how it protects the user.

○ **A.** HTTPS makes the website load faster on public Wi-Fi networks

○ **B.** HTTPS encrypts the data transmitted between the user's browser and the website, so even if attackers on the public Wi-Fi intercept the traffic, they cannot read credit card details or passwords

○ **C.** HTTPS blocks all viruses that might be on the public network

○ **D.** HTTPS prevents other people from using the same Wi-Fi network

Q30 (3 marks)

A gaming café is setting up its network. It will have 20 fixed gaming PCs that require the fastest possible connections for competitive online gaming, and also wants to offer Wi-Fi for customers' personal devices. Evaluate the most appropriate network design.

○ **A.** Use only wireless for everything because cables look untidy in a commercial space

○ **B.** Use only wired connections because wireless is too slow for any gaming

○ **C.** Use wired connections for the gaming PCs (fastest, most reliable, lowest latency) and wireless for customers' personal devices (convenient, flexible)

○ **D.** Use Bluetooth to connect all devices because it is the most

modern technology

Total: **55** marks • Time: **30 min**

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